

1 Background

Semidirect Product $G \ltimes H$: Given groups G, H and a homomorphism $\rho : G \rightarrow \text{Aut}(H)$ (Aut: iso & $V = W$), the semidirect product $G \ltimes H$ is defined as the set $G \times H$ with operation: $(g_1, h_1)(g_2, h_2) = (g_1 g_2, \rho(g_2^{-1})(h_1) h_2)$. If ρ is trivial, $\rho(g) = id_H$, then $G \ltimes H = G \times H$.

Homogeneous Polynomial: $f(x_0, x_1, \dots, x_n) = \sum_i \alpha_{a_1, \dots, a_n} x_0^{a_0} x_1^{a_1} \dots x_n^{a_n}, \sum a_j = d$.
Property: $f(\lambda x_0, \lambda x_1, \dots, \lambda x_n) = \lambda^d f(x_0, x_1, \dots, x_n)$ for any $\lambda \in k$.

2 Plane Curves

2.1 Affine Geometry

Affine Plane $(\mathbb{A}^2, \mathbb{A}_k^2)$: The affine plane over a (ground) field k is defined as $\mathbb{A}_k^2 = k^2$.

Curve: Plane curve $C \subset \mathbb{A}^2$ s.t. $C : p(x, y) = 0$ where p is a non-constant polynomial.

Line (L) : $L \subset \mathbb{A}_k^2 : ax + by + c = 0$ where $a, b, c \in k; (a, b) \neq (0, 0)$

- $\left. \begin{array}{l} L_1 : a_1 x + b_1 y + c_1 = 0 \\ L_2 : a_2 x + b_2 y + c_2 = 0 \end{array} \right\}$ are **equal lines** $\Leftrightarrow \exists d \in k^* \text{ s.t. } a_1 = da_2, b_1 = db_2, c_1 = dc_2$.
- $\left. \begin{array}{l} P_1 = (x_1, y_1) \\ P_2 = (x_2, y_2) \end{array} \right\}$ **distinct points** $\Rightarrow \exists! L : \det \begin{pmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{pmatrix} = 0$ **passes through P_1, P_2** .
- For any $L_1, L_2 \in \mathbb{A}^2$, there are three cases:
 - $L_1 = L_2$
 - $L_1 \cap L_2$ is a point P
 - L_1, L_2 do not intersect (parallel).

Conic: $C \subset \mathbb{A}^2 : ax^2 + bxy + cy^2 + dx + ey + f = 0$ where $(a, b, c) \neq (0, 0, 0)$

- If conic is **reducible**, it's union of 2 lines: $C : (\alpha_1 x + \beta_1 y + \gamma_1)(\alpha_2 x + \beta_2 y + \gamma_2) = 0$
- For a line L and a conic C , there are two cases:
 - C is a union of L and another line L' (C reducible)
 - $L \cap C$ at most 2 points.

Affine Transformation: An affine transformation $\mathbb{A}^2 \rightarrow \mathbb{A}^2$ is given by combinations of:

- Translations:** $(x, y) \mapsto (x + x_0, y + y_0)$ NB: Not Euclidean translations!
- Linear Changes:** $(x, y) \mapsto A(x, y)$ where A invertible 2×2 matrix over k .

Theorem: Group of affine transformations: $\{x \mapsto A(x + b) : A \in GL_2(k), b \in k^2\}$ **isomorphic to semidirect product $GL_2(k) \ltimes k^2$ with operation:** $\rho(A)(b) = Ab$.

- 所有仿射变换可写作 $x \mapsto A(x + b)$.
- $A(x + b) \leftrightarrow (A, b)$ and $(A_1, b_1)(A_2, b_2) = (A_1 A_2, A_2^{-1} b_1 + b_2)$.

2.2 Projective Geometry

$\sim$: Let $\sim$ be e.r. on $k^2 \setminus \{(0, 0)\}$ s.t. $(x, y) \sim (x', y')$ iff $\exists \lambda \in k^*, x' = \lambda x, y' = \lambda y$.

Projective Line $(\mathbb{P}^1)$: $\mathbb{P}^1 = (k^2 \setminus \{(0, 0)\}) / \sim$ (过原点的直线的集合, 无原点).

Topology: $\mathbb{P}_{\mathbb{R}}^1$ is homeomorphic to a circle S^1 . $\mathbb{P}_{\mathbb{C}}^1$ is homeomorphic to a sphere S^2 .

Points $([x : y])$: A point in $\mathbb{P}^1$ is denoted by $[x : y]$, called **homogeneous coordinates**.

- $[x : y] = [\lambda x : \lambda y]$ for any $\lambda \in k^*$.
 - $[0 : 0]$ not defined.
- A point $[a : b] \in \mathbb{P}^1$ specifies a line $L : ax + by = 0$ in k^2 .
 - The map $f : \mathbb{P}^1 \setminus \{[1 : 0]\} \rightarrow k$ by $f([x : y]) = \frac{x}{y}$ is a **bijection**.
 - $P = [1 : 0]$ is called **point at infinity**. (NB: $\mathbb{P}^1 = k \cup \{\infty\}$, " $\infty = P$ "; $\mathbb{P}^1 \setminus P \cong k$).

Projective Space $\mathbb{P}(V)$: Let V be a vector space over k , then $\mathbb{P}(V) = (V \setminus \{0\}) / \sim$

- $\sim : v \sim v' \text{ iff } \exists \lambda \in k^*, v' = \lambda v$.
- $\mathbb{P}^n = \mathbb{P}(k^{n+1})$
- Point $[v]$

Projective Plane $\mathbb{P}^2$: $\mathbb{P}^2 = (k^3 \setminus \{(0, 0, 0)\}) / \sim$ Point in $\mathbb{P}^2$ is $[x : y : z]$.

- Line at infinity:** $U = \{[x : y : 0] : x, y \in k\}$, and $U \cong \mathbb{P}^1$.
- The map $f : \mathbb{P}^2 \setminus U \rightarrow k^2$ by $f([x : y : z]) = (\frac{x}{z}, \frac{y}{z})$ is a **bijection**, and $\mathbb{P}^2 \setminus U \cong \mathbb{A}^2$.

Identification: Affine plane $\mathbb{A}^2$ identified as subset of projective plane $\mathbb{P}^2$ by $z \neq 0$.

Relation between $\mathbb{A}^2$ and $\mathbb{P}^2$: p homogeneous in $\mathbb{P}^2$.

- $C : p(x, y, z) = 0$ in $\mathbb{P}^2 \Rightarrow C \cap \mathbb{A}^2$ will in $\mathbb{A}^2$ by $p(x, y, 1) = 0$. (**dehomogenization**)
- $C : p(x, y) = 0$ in $\mathbb{A}^2 \Rightarrow \tilde{C} : z^d p(\frac{x}{z}, \frac{y}{z}) = 0$ in $\mathbb{P}^2$ (**homogenization**)

2.3 Projective Curves

Projective Curve: $C \subset \mathbb{P}^2, C : p(x, y, z) = 0$ where p homogeneous polynomial.

ps: Homogeneous ensure that: $p(x, y, z) = 0 \Rightarrow p(\lambda x, \lambda y, \lambda z) = 0, \forall \lambda \in k^*$.

Line (L) : $L \subset \mathbb{P}^2, L : ax + by + cz = 0$ where $(a, b, c) \neq (0, 0, 0)$.

- $\left. \begin{array}{l} L_1 : a_1 x + b_1 y + c_1 z = 0 \\ L_2 : a_2 x + b_2 y + c_2 z = 0 \end{array} \right\}$ are **equal lines** $\Leftrightarrow \exists d \in k^* \text{ s.t. } a_1 = da_2, b_1 = db_2, c_1 = dc_2$.
- $\left. \begin{array}{l} P_1 = [x_1 : y_1 : z_1] \\ P_2 = [x_2 : y_2 : z_2] \end{array} \right\}$ **distinct** $\Rightarrow \exists! L : \det \begin{pmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{pmatrix} = 0$ **passes through P_1, P_2** .
- For any $L_1, L_2 \in \mathbb{P}^2$, there are two cases:
 - $L_1 = L_2$
 - $L_1 \cap L_2$ is a point P .

Conic: $C \subset \mathbb{P}^2, C : ax^2 + bxy + cy^2 + dxz + eyz + fz^2 = 0$ where $(a, \dots, f) \neq (0, \dots, 0)$

- If conic is **reducible**, it's union of 2 lines: $C : (\alpha_1 x + \beta_1 y + \gamma_1 z)(\alpha_2 x + \beta_2 y + \gamma_2 z) = 0$
- For a Line L and a Conic C , there are two cases:
 - C is a union of L and another line L' (C reducible)
 - $L \cap C$ at most 2 points.
- Any 5 distinct points $P_1, \dots, P_5$ no 3 collinear, $\exists!$ conic C passing through them.**

Projective Transformation: It's a map $\phi : \mathbb{P}(V) \rightarrow \mathbb{P}(V)$ by $\phi([v]) = [T(v)], T \in GL(V)$.

Property: $\text{PGL}(V) = \text{GL}(V) / k^*, k^* = \{\lambda \cdot id_V : \lambda \in k^*\}$. i.e. If $T_1 = \lambda T_2$, 对应射影变换相同.

- For line: $\forall L \subset \mathbb{P}^2, \exists$ projective transformation ϕ s.t. $\phi(L) : x = 0$.
- For points: **Given 4 points P_1, P_2, P_3, P_4 no 3 collinear**, $\exists$ projective transformation ϕ s.t. $\phi(P_1) = [1 : 0 : 0], \phi(P_2) = [0 : 1 : 0], \phi(P_3) = [0 : 0 : 1], \phi(P_4) = [1 : 1 : 1]$.